Unit: 6 Unit 8: Quadratics
Topic: Systems of Linear and Quadratic Equations

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. Solve the system: $x + y = 3$, $x^2 + y^2 = 9$
(A) (0, 3)
(B) (3, 0)
(C) (1, 2)
(D) (-1, 2)
(E) No solution

Q2. Find the intersection point of $y = 2x - 1$ and $y = x^2 - 4$
(A) (-1, -3)
(B) (2, 3)
(C) (-2, -3)
(D) (1, -1)
(E) (3, 5)

Q3. Solve the system: $y = x^2 - 2$, $y = -x + 1$
(A) (1, -1)
(B) (-1, -1)
(C) (0, -2)
(D) (2, 2)
(E) No solution

Q4. Find the intersection point of $x - y = 2$ and $x^2 + y^2 = 8$
(A) (2, 0)
(B) (0, -2)
(C) (-2, 0)
(D) (2, 2)
(E) (-1, 3)

Q5. Solve the system: $y = x^2 + 1$, $y = -2x - 1$
(A) (-1, 2)
(B) (1, 2)
(C) (-2, 3)
(D) (0, 1)
(E) No solution

Q6. Find the intersection point of $2x + y = 4$ and $y = x^2 - 3$
(A) (1, 1)
(B) (-1, 5)
(C) (2, 0)
(D) (-2, 0)
(E) (0, 4)

Q7. Solve the system: $x^2 - y^2 = 1$, $x + y = 3$
(A) (2, 1)
(B) (-2, -1)
(C) (1, 2)
(D) (-1, -2)
(E) (1, -1)

Q8. Find the intersection point of $y = x^2 - 2x - 3$ and $y = x - 1$
(A) (-1, -2)
(B) (3, 2)
(C) (1, 0)
(D) (-2, 1)
(E) (2, 1)

Open Ended Questions (FRQ)

Q9. Solve the system: $x^2 + y^2 = 16$, $x - y = 2$. Find the intersection points and show your work.

Q10. Find the intersection points of $y = x^2 - 4x + 3$ and $y = 2x - 3$. Show your work and explain your reasoning.

Chef's Tips

- Provide clear and concise explanations for each question.
- Encourage students to show their work and explain their reasoning.
- Use the points of intersection to interpret the results.